



KOLEJ MATRIKULASI NEGERI SEMBILAN

EXAM STUDY HACKS

SET 1

PROGRAM KECEMERLANGAN AKADEMIK UNIT FIZIK
MINGGU KE 13 - 15

FIZIK (DP024)

2 jam

SET SOALAN DARIPADA KOLEJ MATRIKULASI

NO	MARKS
1	/16
2	/12
3	/10
4	/14
5	/14
6	/14
TOTAL	/80

Answer **ALL** questions.

1. a) The progressive wave with amplitude 2.0 cm, frequency 10 Hz and wavelength 12.8 cm is moving to the left. Calculate
- period.
 - progressive wave equation.
- [5 marks]
- (b) A progressive wave can be represented by the equation $y(x,t) = 20 \sin(8\pi t - 6x)$ where y and x are in centimeter and t in seconds. Determine .
- direction of wave propagation
 - amplitude
 - frequency
 - wavelength
 - wave propagation velocity
 - particle vibrational velocity at $x = 3$ cm and $t = 3$ s
- [11 marks]
2. a) A copper wire has a diameter of 0.395 mm with resistance is 1.10Ω . Given the resistivity of copper is $1.72 \times 10^{-8} \text{ N m}$.
- Calculate the length of the copper wire.
 - If a current of 2 A flows through the wire within 1 minute, calculate the quantity of charge that pass through the wire during that time.
- [4 marks]

b)

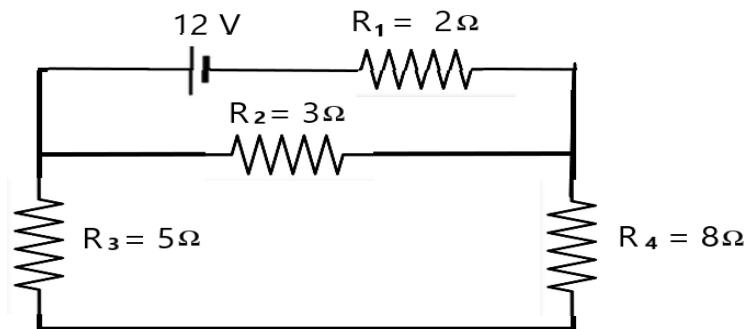


FIGURE 2

Four resistors are connected to a battery as shown in FIGURE 2. Calculate the

- effective resistance of the circuit.
- total current in the circuit
- voltage across resistor R_1

[8 marks]

3. (a)

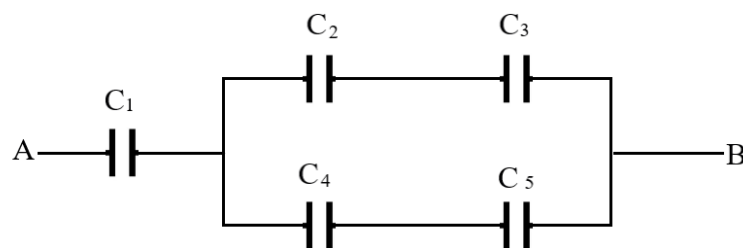


FIGURE 3a

FIGURE 3a shows five (5) identical capacitors of $8\ \mu\text{F}$ connected between point A and B. If the voltage supply across point A and B is $12\ \text{V}$, calculate the

- (i) effective capacitance.
- (ii) total charge in the circuit

[4 marks]

b)

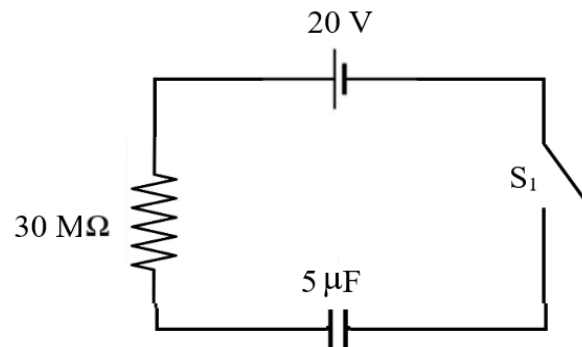


FIGURE 3b

FIGURE 3b shows a direct current circuit. When switch S_1 is closed, the capacitor is charging,

- (i) calculate the time constant. hitungkan pemalar masa.
- (ii) calculate the time taken for the charge to reach 25% of its maximum
- (iii) sketch a graph of current against time

[6 marks]

4. a)



FIGURE 4a

FIGURE 4a shows the top view of two straight current-carrying conductor. Sketch the magnetic field lines produced by the two conductors.

[2 marks]

b) A solenoid has a length of $0.275\ \text{m}$, radius $2.4\ \text{cm}$ and 215 number of turns. Calculate:

- (i) magnetic field at the centre of the solenoid.
- (ii) maximum magnetic flux produced by the solenoid.

[4 marks]

c)

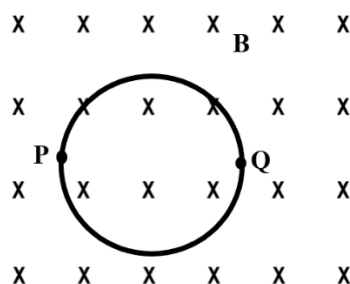


FIGURE 4c

FIGURE 4c shows a flexible single loop of radius 15.0 cm situated perpendicularly in a magnetic field of 0.125 T. The loop is being stretched by grasping point P and Q until its area is almost zero

- (i) If the time taken is 0.3 s, calculate the magnitude of induced emf in the loop.
- (ii) If the loop is replaced with a rigid single loop of a resistance 0.5 Ω , with the same radius and the magnetic field decreases at the rate of 0.04 T/s, calculate the magnitude of the induced emf.
- (iii) Refer to (c)(ii), calculate the current in the loop.
- (iv) By using Lenz's law, state the direction of induced current in (c)(iii).

[8 marks]

5. a) An upright image is formed 20.5 cm from the real object by using the spherical mirror. The image's height is one fourth of object's height.

- (i) Calculate the image distance from the mirror.
- (ii) Calculate the radius of curvature of the mirror and state the type of mirror is used.
- (iii) Sketch and label a ray diagram to show the formation of the image.

[8 marks]

- b) An object is placed 15 cm in front of a convex lens with focal length 10 cm.

- (i) Calculate the image distance and magnification of the image.
- (ii) State two (2) characteristics of the image.

[6 marks]

6. a) In a double-slits experiment, the distance between the two slits is 0.25 mm. The third bright fringe is 7.45 mm from the central bright. If the distance from a double-slits to the screen is 2.1 m,

- (i) calculate the wavelength of the light.
- (ii) calculate the distance of the third dark fringe from the central bright.
- (iii) calculate the distance between two consecutive bright fringes.
- (iv) state the effect of the fringes separation if the separation distance (AL) between double slit is decreased. Explain your answer.

[10 marks]

- b) A mixture of red light (wavelength = 665 nm) and yellow green light (wavelength = 565 nm) incident on the double-slits of separation 2 mm. A flat screen is located 2.25 away from double slits. Calculate the

- (i) distance of third order red fringe from the central bright.
- (ii) distance of fourth order yellow green fringe from the central bright.
- (iii) separation distance between third order red fringe and fourth order yellow green fringe.

[4 marks]

END OF QUESTION PAPER

ANSWERS:

- 1.(a) (i) 0.1 s (ii) $y(x,t) = 2.0 \sin (62.83t + 0.491x)$
 (b) (i) To the right (ii) 20 cm (iii) 4 Hz (iv) 1.047 cm (v) $0.042 \text{ ms}^{-1}/4.20 \text{ cm s}^{-1}$
 (vi) $3331.91 \text{ cm s}^{-1}/3.3191 \text{ ms}^{-1}$
- 2.(a) (i) 7.84 m (ii) 120 Ω (b) (i) 4.44 Ω (ii) 2.70 A (iii) 5.40 V
- 3.(a) (i) $4\mu\text{F}$ (ii) $48\mu\text{C}$ (b) (i) 150 s (ii) 43.15 s (iii) DIY
- 4.(a) DIY (b) (i) $3.14 \times 10^{-3}\text{T}$ (ii) $5.68 \times 10^{-6}\text{Wb}$
 (c) (i) 0.029 V (ii) $2.83 \times 10^{-3}\text{V}$ (iii) $5.66 \times 10^{-3}\text{A}$ (iv) Clockwise
- 5.(a) (i) 4.10 cm (ii) -10.94 cm Convex mirror (iii) DIY
 (b) (i) -2 (ii) Real, inverted and magnified (RIM)
- 6.(a) (i) $2.96 \times 10^{-7}\text{m}$ (ii) $6.22 \times 10^{-3}\text{m}$ (iii) $2.49 \times 10^{-3}\text{m}$ (iv) increase Δy inversely d
 (b) (i) $2.24 \times 10^{-3}\text{m}$ (ii) $2.54 \times 10^{-3}\text{m}$ (iii) $3.04 \times 10^{-4}\text{m}$